SUBMISSION TO THE

National Strategic Risk Assessment 2026

Public Consultation · Department of the Taoiseach

Submitted by: Brøek Meinhardt, Independent Researcher

Subject: Analytical capability, cross-departmental coordination and national resilience

Date: October 9th, 2026

Summary

The principal risk this submission identifies is one of absorptive capacity: an apparent incapacity, at the level of the State as a whole, to receive, evaluate and institutionally incorporate a body of Operations Research methods, and the ready-to-use technology that applies them, notwithstanding their having been placed before Government repeatedly, and through a plurality of channels, across the past two decades.

Since 2006 the author has laid this material before State bodies; its strategic dimensions were placed before the Taoiseach's Office at the outset, and the collated record now comprises thirteen reference-numbered acknowledgements from six Departments since June 2024 alone (Part III). None has issued in a determination, and the record assembled for this submission identifies no office, committee or statutory function that holds the anterior question: whether the State possesses, or requires, a standing capacity to integrate analytical offers of this kind.

This bears on more than the author's own correspondence, because every other risk in the Assessment is, in some measure, hostage to it. A State may identify a risk with precision and remain unable to model how several such risks interact, or to allocate scarce resources rationally when they converge. Operations Research is the discipline that supplies that capacity. It finds no mention in the impact provisions of the Cabinet Handbook, and S.I. No. 559 of 2024 requires interdependency analysis while prescribing no method by which it is to be performed. The deficiency lies in procedure, not in personnel: it is not a finding on the discipline's merit, and no fault is imputed to any office-holder.

The same structural deficiency does not terminate at the State's own departmental boundaries. The instruments by which Ireland coordinates with the United Kingdom and with Northern Ireland, the North/South Ministerial Council, the British-Irish Council, the all-island Single Electricity Market, the Windsor Framework's arrangements for the movement of goods, rest on the identical premise that a capacity exists somewhere to model how a disturbance in one domain propagates through the others. The public record located for this submission does not show that it does. An eighth proposed risk, introduced below, extends the analysis to that Anglo-Irish and cross-border dimension (Part IX.H).

Ireland has recently built such a coordinating capacity for artificial intelligence (the AI Office of Ireland, established under the Regulation of Artificial Intelligence Act 2026 and now co-located within the Department of Enterprise, Tourism and Employment with the dual-use export-control function). The technical demands of the AI risk-management lifecycle overlap substantially with established Operations

Research methods. The institutional answer given for one domain has not yet been given for the other —even though the two domains are intrinsically linked and, indeed, inseparable in some cases.

The submission proposes eight related risks. The first, insufficient integrated analytical and decision-support capability, is the principal one. Risks 2 to 5 describe the links by which the State recognises signals, assembles competences, converges on a decision and corrects its errors. Risk 6 shows the same gap in digital infrastructure, Risk 7 shows it in the staged procurement of a critical operational system, and Risk 8 shows it across the Anglo-Irish and North-South boundary. Each has a named discipline, and all of them belong to Operations Research.

The submission does not ask the Government to adopt any method, technology or supplier, and it does not propose an institutional form. It asks the assessment to establish where the capability sits and how its competences are assembled (Part X), and to record the answer.

Responses to the Department's Questions

Q1. Existing risks. The twenty-four risks identified in 2024 remain relevant, and none should be removed. Risk 20 (Artificial Intelligence and other advanced technologies) should be amended to record that integrated analytical capability is a cross-cutting dependency of every other listed risk, not a technological risk alone. Risk 1 (Increase in global insecurity and division) should be framed as a structural transition of the international order and not only as episodic insecurity: in the World Economic Forum's *Global Risks Report 2026*, 68% of respondents expect a multipolar or fragmented order over the next decade, up four points on the previous year, and geoeconomic confrontation ranks as the top risk for 2026. A State that prepares for episodic shocks and not for a changed order will mis-specify the resilience it needs. Risk 9 (Protectionism and deglobalisation) should distinguish temporary disruption from structurally higher trade friction, and should extend to services trade. A structural reading changes what Ireland prepares for: compliance, traceability and origin-verification burdens fall on exporters and on the State bodies that certify them, and they persist after any single tariff measure is reversed.

Risk 21 (Energy security) should extend beyond import dependence and electricity adequacy to the domestic logistics of fuel: the refinery, depots, ports and road distribution on which supply depends, and their exposure to disruption other than supply shortfall. The blockades that began on April 7th, 2026 at the Whitegate refinery and the depots at Galway and Foynes put access to roughly half of national fuel at risk within days, according to the industry body Fuels for Ireland. More than 100 forecourts had run dry by April 10th, 2026, and the industry body later reported around 600. The National Emergency Coordination Group was convened, and the event engaged Transport, Energy, Justice, Defence and Finance at once.

Q2. New risks. Eight related risks are proposed. Risk 1 is the principal one: the State's inability to integrate analytical and decision-support capability, including the methods and technology offered to it over twenty years. Risks 2 to 5 describe the State's capacity to recognise, understand, integrate and act on knowledge about the risks already listed, and to detect when it has failed to do so. Risk 6 applies the same concern to a concrete asset class, the digital infrastructure on which several listed risks converge. Risk 7 applies it to a concrete delivery case, the staged procurement and funding of a critical operational system. Risk 8 applies it across a jurisdictional boundary, the Anglo-Irish and North-South coordination architecture through which several of the same listed risks are, in part, already managed. None replaces an existing risk. Each is a cross-cutting dependency of several of them (Parts VIII and IX).

#	Proposed risk	Category, if the five-category structure is retained	Cross-reference	Method class whose absence it names
1	Insufficient integrated analytical and decision-support capability (principal risk; Part VIII)	Technological	All categories	Prescriptive analytics; optimisation; simulation
2	Epistemic failure: signals that do not reach determination	Societal	Technological	Early-warning, sequential and anomaly detection; forecast monitoring
3	Institutional non-integration	Societal	Technological; Geopolitical	Decomposition and hierarchical planning
4	Convergent responsibility and executive bottlenecks	Societal	Geopolitical	Queueing and bottleneck analysis
5	Loss of institutional self-correction	Societal	All categories	Statistical process control; earned-value feedback
6	Geoeconomic exposure of digital infrastructure (Part IX.F)	Technological	Economic; Geopolitical	Network, infrastructure and dependency analysis
7	Delivery of critical national systems (Part IX.G)	Economic	Technological; Societal	Stage-dependency and critical-path analysis; earned-value management; real options
8	Fragmented Anglo-Irish and cross-border coordination capacity (Part IX.H)	Geopolitical	Societal; Economic	Multi-agent and decentralised coordination; cross-boundary network and dependency analysis

Every entry in the right-hand column belongs to Operations Research. The eight risks are eight views of one missing capability, and Risk 1 is the one on which the other seven depend.

Risks 2 to 5 could form a sixth category, *Institutional and Governance Risks*. If the five-category structure is retained, they sit most naturally under Societal Risks, alongside social cohesion, with cross-references to the others.

Q3 to Q5. Private individual; organisation type not applicable; based in Ireland.

I. The capability question

Operations Research, the discipline that structures decisions about interacting risks and constrained resources, has no procedural entry point in the machinery of government. The exclusion is procedural. It is not a judgement on the discipline's merit, which is not in question: Operations Research is an established analytical discipline, originating in military planning and now routinely embedded internationally in defence governance, logistics, emergency management and high-consequence decision-making. NATO and the defence establishments of the United Kingdom and the United States, together with a number of EU Member States, maintain standing operational-analysis functions within their defence governance. The absence of an equivalent entry point in the Irish machinery of government is therefore a departure from the international pattern and not a reflection of it.

A State can identify a risk correctly and still lack the means to decide what to do when several risks arrive together. The NSRA identifies risks. The capacity to analyse how they interact, to evaluate alternative courses of action, and to allocate scarce resources across competing systems under simultaneous stress is a separate capacity, and, on the public record, a separate gap.

The discipline was itself an answer to an integration problem. In Britain in the late 1930s, operational research was applied to join radar, fighter control and command into a single system of air defence whose components already existed. Its founding insight was that a system of functioning parts is a distinct object of analysis. The present question is that insight, applied to the State.

This submission places that capacity before the 2026 assessment as a cross-cutting strategic risk, and advances one proposition: the capacity is a coordination problem before it is a technology problem. The competences it requires sit in different Departments, under different statutes, answerable to different Ministers. The Constitution makes the Government collectively responsible for all of them (Article 28.4.2°). What remains open is whether the State holds a standing arrangement that assembles those competences into one analysis before a decision is taken, or relies on consultation after a proposal has taken shape, which is what the Cabinet Handbook requires at ¶3.8.

> *The proposition has stayed the same while the technology has grown, and the architecture that would connect them is distributed by design.*

II. Twenty years of continuity, one change of technology

Since 2006 the author has presented Operations Research material and applications to State bodies, in connection with emergency logistics, humanitarian operations, health-system pressures and Defence-relevant functions, and the strategic dimensions were placed before the Taoiseach's Office at the outset. The author is the person who made those submissions. The interest is declared so that the Department may weigh the evidence accordingly. No consolidated, documented technical assessment of the material has been identified on the departmental record.

The proposition has stayed recognisable through changes of Government, departmental allocation and terminology: public administration requires the capacity to model complex systems, represent interacting constraints, forecast demand and capacity, evaluate alternatives, optimise constrained resources, run scenario and contingency analysis, identify bottlenecks and cascading effects, and support decisions under uncertainty.

Technology is what has changed. Computing, data infrastructure and artificial intelligence have widened what can practically be done; they have not altered the question. Three questions should be kept apart: what the risk is, what analytical capability is needed to understand and manage it, and what technology can supply that capability. AI belongs to the third. The second is the strategic issue, and the development of AI does not retrospectively turn the 2006 proposition into an AI proposal.

AI is not a substitute for the disciplines that structure the problem. Operations Research supplies established methods for defining objectives, constraints, decision variables, feasible alternatives, trade-offs and performance measures; AI extends the computational setting in which those methods operate. Where decisions are high-consequence and explainability, validation, data provenance, human oversight and accountability matter, the opportunity is an integrated capability, with Operations Research, decision science and AI operating together, not AI in place of the discipline.

III. Thirteen acknowledgements, no determination: the risk realised

The proposition has not existed solely outside Government. The collated record contains thirteen reference-numbered acknowledgements from six Departments since June 2024. Read together, they are a chronology of the risk described in Part IV after it has occurred.

Date	Reference	Department
June 19th, 2024	DOT-TH-03856-2024	Department of the Taoiseach
August 22nd, 2024	FIN-MO-03632-2024	Department of Finance
August 23rd, 2024	DFA-TMO-05256-2024	Department of Foreign Affairs
September 4th, 2024	DOD-MO-01206-2024	Department of Defence
October 4th, 2024	DOT-TH-11219-2024	Department of the Taoiseach
February 10th, 2025	DOT-TM25-01179-2025	Department of the Taoiseach
May 16th, 2025	DOT-TM25-06516-2025	Department of the Taoiseach
July 3rd, 2025	DOD-MO-00707-2025	Department of Defence
September 5th, 2025	DFAT-THO-04011-2025	Department of Foreign Affairs and Trade
December 15th, 2025	DOD-MO-01201-2025	Department of Defence
April 8th, 2026	TTAS-MO-01932-2026	Department of Transport
May 28th, 2026	DETE-MBURK-01636-2026	Department of Enterprise, Tourism and Employment
September 10th, 2026	DETE-MBURK-02755-2026	Department of Enterprise, Tourism and Employment

An acknowledgement is not an endorsement, receipt is not assessment, and institutional contact is not determination. The evidential proposition is narrower, and it holds: the strategic proposition has repeatedly entered governmental attention across successive departmental and governmental configurations. Departments have responded on the use of particular methods within their own sectors. Those responses answer a sectoral question. The record located for this submission contains no response to the cross-departmental one: whether the State has a pathway that converts continuing awareness into an assessment of national capability. On that record, none of the thirteen identifies a body that holds the question. The discipline has a vocabulary for the pattern. Thirteen arrivals in twenty-seven months, roughly one every two months, and no departures: in queueing terms, an arrival process with no service completions, in which the queue grows without bound. In the terms of earned-value management, the record shows activity, thirteen acknowledgements, and no earned value, since earned value is work accomplished measured against plan and no determination was produced to measure. The terms are offered descriptively. They are the terms in which an analyst would describe the record, and the discipline that supplies them is the one under discussion.

IV. The risk of non-coordination

Part III records the risk after it has occurred. The risk itself is that a matter bearing on several Departments has no point at which their competences are assembled into one analysis.

The executive recognises the problem for its own business. Under the Cabinet Handbook, a Minister with a functional interest in a proposal must be heard, and views not accepted are recorded in the memorandum

(¶3.8). That is consultation. It gathers opinions on a proposal once formed. It does not require that anyone has modelled how the risks the proposal touches interact, and no impact heading asks for it (¶3.4; Appendix III). No breach is alleged. The Handbook describes itself as guidance that the Government may amend or disapply, and it secures what it was written to secure.

On March 3rd, 2026, Senator Michael McDowell told the Seanad Select Committee on EU Scrutiny and Transparency that this forewarning "is not replicated in the Oireachtas between the various committees." The remark concerned EU legislative scrutiny and is recorded here as the Senator's. Its relevance is structural: the executive has a consultation step, the committee system has none, and neither has an analytical one.

Consultation gathers opinions on a proposal. Analysis assembles the competences needed to model it. Operations Research is the discipline that performs the second, and it has no named place in either procedure.

V. The statutory architecture: distributed by design

A. The general statute. Section 4(1)(e) of the Public Service Management Act 1997 requires each Secretary General to ensure that appropriate arrangements are in place to facilitate an effective response to matters that pertain both to the Department and to other branches of the public service. Section 12 allows Ministers, jointly and after consulting the Secretary General of each Department concerned, to assign civil servants by order responsibility for functions relating to matters that pertain to more than one Department; the order must state its own cessation date, and the Ministers keep concurrent power to act (s. 12(3), (5)). Together the provisions form the statutory floor: a duty at Secretary General level to arrange cooperation, and an optional instrument at Ministerial level to assign cross-departmental functions. Neither establishes an analytical function, and the second is permissive. The Taoiseach told Dáil Éireann on April 27th, 1999 that no order had been made under section 12; none has been located in the sources consulted for this submission.

B. The administrative layer. The Government describes the Taoiseach as the central co-ordinator of the work of Ministers and their Departments, a description of role and not a statutory power. For emergencies, coordination runs through the Government Task Force on Emergency Planning, chaired by the Minister for Defence and established in October 2001, supported by the Office of Emergency Planning, created by Government decision in the same month. The architecture operates on a lead-Department principle: the Department responsible for an activity in normal conditions keeps that responsibility in a national emergency, and the Office convenes the National Emergency Coordination Group within one hour of notification by a lead Department. Each element is a creature of Government decision. Whether the Office should have a statutory footing was put to the Minister in the Dáil on March 25th, 2004.

C. The first statutory anchor. S.I. No. 559 of 2024, the European Union (Resilience of Critical Entities) Regulations 2024, made by the Minister for Defence under section 3 of the European Communities Act 1972 to transpose Directive (EU) 2022/2557 and in operation from October 17th, 2024, gives the Minister for Defence statutory functions the Task Force architecture lacked. The Minister prepares the National Risk Assessment of Ireland (regs 2 and 11) and the National Strategy on the Resilience of Critical Entities (reg. 10), and acts as a single point of contact, liaising with each competent authority and empowered to require information from them (reg. 7). Regulation 8(2) designates thirteen competent authorities across the sectors in the Schedule, from the Commission for Regulation of Utilities to the Minister for Agriculture, Food and the Marine; regulation 9(5) requires each to participate in and inform a national risk assessment.

D. What the instrument requires, and what it leaves open. The national risk assessment must account for risks of a cross-sectoral or cross-border nature (reg. 11(3)) and for the extent to which sectors depend on one

another, including the effect of a significant disruption in one sector on others (reg. 11(4)(c)). The Strategy must address cross-sectoral dependencies and interdependencies and set out a governance framework describing the roles of the different authorities (reg. 10(2)(a), (b)). The statute therefore requires interdependency analysis and names the Minister who must carry it out. It prescribes no method, and it does not say where the modelling capacity sits. That capacity is presupposed.

E. The EU basis. The national risk assessment is also the report Ireland owes under Article 6 of Decision No 1313/2013/EU on a Union Civil Protection Mechanism, which treats risk assessment as an overall cross-sectoral process of identification, analysis and evaluation (Article 4) and requires summaries to the Commission at three-year intervals. The Minister for Defence published the National Risk Assessment for Ireland 2023 on March 28th, 2024, forwarded it to the Commission under Article 6, and stated in the Dáil on May 16th, 2024 that officials in the Department had begun a cross-government process to assess Departments' capabilities against the 22 national-level risks it identified.

VI. Three risk exercises, no standing interaction model

The Department's own consultation document names three exercises: the NSRA itself, led from the Department of the Taoiseach since 2014 and expressly not a substitute for Departmental risk management; the National Risk Assessment for Ireland 2023, an EU-required assessment of natural, transportation, technological and civil emergency risks led by the Department of Defence; and the National Cyber Risk Assessment of the National Cyber Security Centre. The Government's release of March 28th, 2024 describes the second as complementing the first. Three exercises, three owners, one risk landscape. In the language of large-scale optimisation, these are three independent solutions to a coupled problem, with no master problem to reconcile them.

Public descriptions of the three describe identification, rating and horizon scanning. The 2023 assessment publishes a national risk matrix of 22 key risks and an associated research project, FUTUREPROOF-IE, on horizon scanning and emergent-risk forecasting. The public record located for this submission does not describe a standing capability to model how the listed risks interact and to optimise the allocation of scarce resources across them. Whether such a capability exists elsewhere in Government is the fact the NSRA is positioned to establish. Identification, rating and horizon scanning correspond to the descriptive and predictive tiers of analytics. Part VIII returns to the third.

Two existing structures show where it could attach. The Defence-led process of assessing Departmental capabilities, begun in 2024, is the nearest analogue to the capability question, and the National Risk Assessment is approved by the Government Task Force on Emergency Planning, on which Ministers or senior officials from all Departments sit. The Department of the Taoiseach also hosts the National Security Analysis Centre, established in 2019 to coordinate between State bodies with national security functions and to provide strategic analysis to Government on security threats. A standing, cross-departmental analytical function therefore exists for threat assessment inside this Department. Its counterpart for the interaction of risks and the allocation of resources is the open item.

VII. What the State has already shown it can build

Ireland's implementation of the EU Artificial Intelligence Act supplies a recent specimen of the design logic. Following Government decisions of 4 March 2025 and 22 July 2025, the Regulation of Artificial Intelligence Act 2026, signed on 21 July 2026, established Oifig IS na hÉireann (the AI Office of Ireland) as an

independent statutory body acting as central coordinating authority within a distributed model of sectoral authorities. Both the AI Office and the competent-authority function for dual-use items under Regulation (EU) 2021/821 and the Control of Exports Act 2023 now sit within the Department of Enterprise, Tourism and Employment under the same ministerial delegation structure. This co-location is material, not incidental —see DETE-MBURK-01636-2026 and DETE-MBURK-02755-2026.

The General Scheme published in February 2026 described the design in terms that bear directly on this submission: a single point of contact, a statutory Cooperation Forum meeting at least quarterly, a centralised pool of technical, legal and regulatory expertise available to all competent authorities, and accountability of the Chief Executive to the Public Accounts Committee (Heads 9, 15 and 34). The Act largely followed the Scheme, with one notable change: the Office coordinates and serves as a single point of contact but is not itself a market surveillance authority.

Herbert A. Simon —one of the founders of artificial intelligence, Nobel laureate in Economics and architect of bounded rationality— described the persistent distinction between Operations Research and AI as historically contingent and institutionally artificial. In 1987 he wrote:

> *"But it is preferable to stress continuities rather than differences. Instead of differentiating between OR and AI, we need to confuse, blend, and synthesize them as much as possible."*

and further:

> *"We need to build our professional institutions and organizations to use them together, supporting, reinforcing, and extending each other."*

— *Interfaces* 17(4), 1987

The two are sister subjects —a fact not explicitly present in any AI-facing regulation. The technical content of Article 9 of the AI Act (risk management system) falls substantially within the established analytical domain of Operations Research: lifecycle risk identification, estimation, evaluation, optimisation of mitigation measures, testing against defined metrics and probabilistic thresholds, and iterative reassessment. The Regulation does not prescribe Operations Research as a methodology; it does, however, prescribe a class of analytical activity in which Operations Research is an established technical discipline. The specific tools that process applies —fault-tree analysis, Monte Carlo simulation, extreme-value analysis, reliability modelling and condition-based optimisation— are well-established within Operations Research and its sub-domains.

Three limits apply to the comparison. The Office coordinates; it does not assemble a single decision from distributed competences. Cooperation agreements under the Scheme were permissive. And the comparison concerns institutional design only: no claim is made that the AI Act regulates Operations Research.

Within those limits the relevance is evidential. In the domain of one regulation the State has defined what a functioning coordination architecture consists of (a convenor, a standing forum, pooled expertise, accountability to an Oireachtas committee) and legislated for it within two years of the regulation entering into force. Ireland has answered the institutional question affirmatively for AI. The export-control record and the twenty-year Operations Research corpus present the inverse problem: DETE, the Department of Foreign Affairs and Trade and the Department of Defence each hold part of the information or institutional function necessary to assess the material, yet the public record identifies no equivalent mechanism by which those inputs are assembled into a competent-authority determination.

The State has built a coordinating office for one computational technology (AI). The discipline that supplies that technology's constraints, validation methods and performance guarantees has no comparable office, even

though the technical demands of the two regimes substantially overlap and both functions now sit within the same Department.

VIII. The principal risk

Insufficient Institutional Knowledge. Operations Research is a quantitatively demanding graduate-level discipline. Its core methods (mathematical programming, queueing theory, stochastic simulation, network optimisation and decision analysis, set out in the table below) rest on linear algebra, probability and optimisation theory, and its practitioners are drawn predominantly from the mathematical, physical and engineering sciences. Defence establishments staff it accordingly. NATO, the United Kingdom, the United States and a number of EU Member States maintain standing operational-analysis functions (Part I). The UK Government Operational Research Service is a professional cadre of more than 1,000 analysts, and it recruits from mathematics, statistics, physics, engineering and other numerate disciplines. The US Army selects its operations research and systems analysis officers for a strong mathematical background and trains them in calculus, statistics, probability, modelling, simulation and decision analysis; many of its posts require a master's degree or doctorate. In each case the analytical function is a distinct career path with its own entry standard, and general policy staff do not absorb it as an ancillary duty.

The test-taker data published by the Educational Testing Service (ETS) show the same quantitative profile (*Figure 1*). Among seniors and non-enrolled college graduates who took the GRE General Test between July 1st, 2022 and June 30th, 2025 and named Operations Research as their intended graduate major (436 test takers), the mean Quantitative Reasoning score was 163 on a scale of 130 to 170. About 72% scored 160 or above, and about 53% scored 165 or above. Of the majors shown in Figure 1 with at least 250 test takers, only Mathematics (164), Applied Mathematics (165) and Statistics (165) record a higher mean, and Electrical, Electronics and Communications Engineering records the same (163). The figure reports test-taker scores by intended major, as ETS publishes them. It describes the quantitative preparation on which the field draws and makes no claim about any other attribute of the people who study it. A State that lacks a standing pathway for the evaluation and institutional incorporation of such material therefore confronts a particular form of absorptive-capacity risk. The ordinary channels of departmental assessment are built for policy, legal and administrative questions. A submission that requires specialist quantitative review can remain outside them for that reason alone, and the more specialised the offer, the more easily it does.

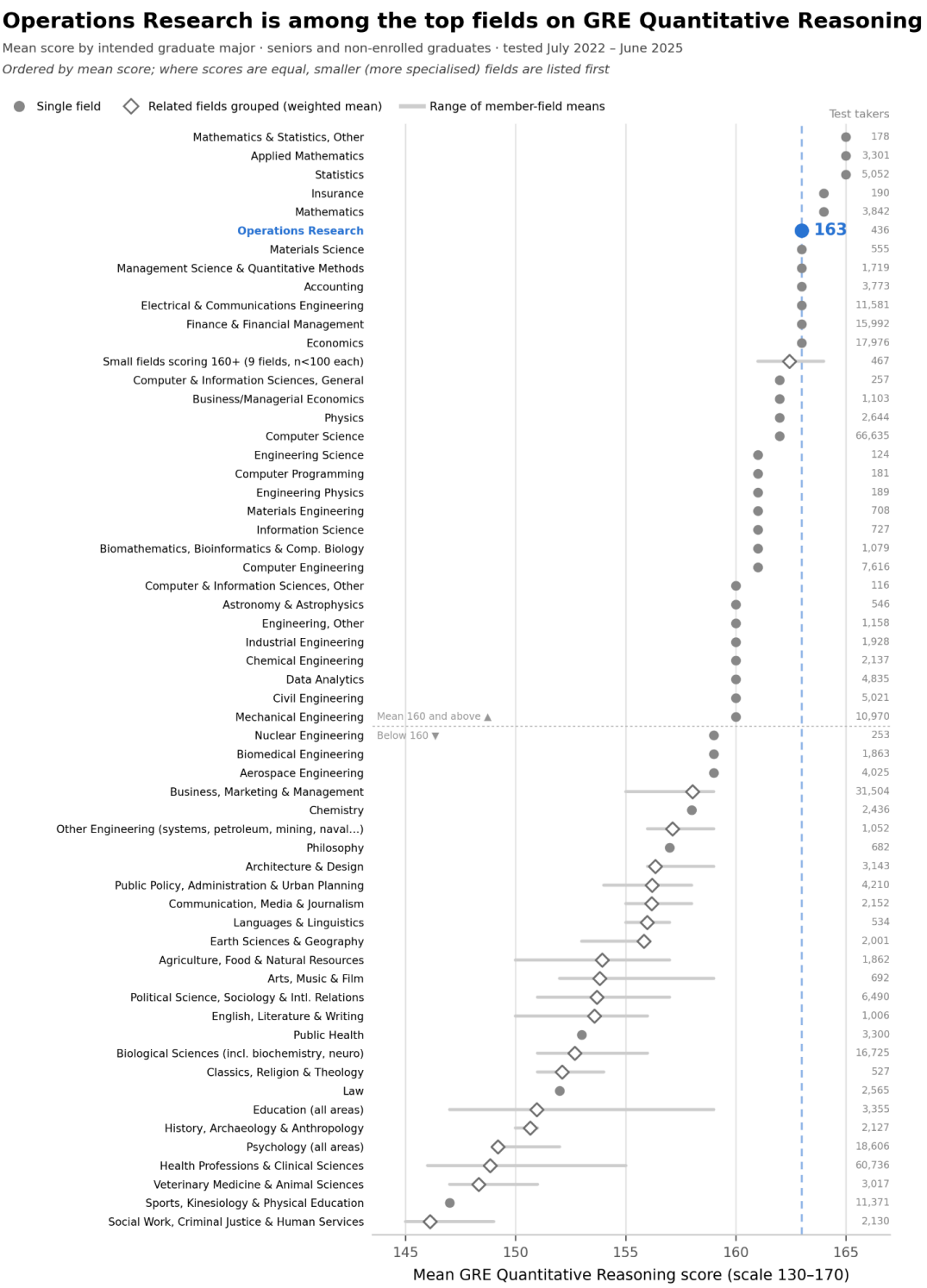


Figure 1. Mean GRE Quantitative Reasoning score of test takers, by intended graduate major. Source: Educational Testing Service, GRE Guide to the Use of Scores, Table 3B (specific majors) and Table 3A (broad fields); seniors and non-enrolled college graduates who tested between July 1st, 2022 and June 30th, 2025. ETS publishes means rounded to whole numbers. Filled dots are Table 3B majors with at least 250 test takers, drawn from the table's groups running from Agricultural Sciences to Theology; "Other" and general-studies rows are omitted. Open diamonds are Table 3A broad fields, shown for the fields that follow those groups in the table (Physical Sciences, Psychology, Social Sciences, Business, Health Professions and others). The figure reports test-taker scores by intended major only. Chart prepared by the author from the published tables.

Insufficient integrated analytical and decision-support capability. The risk is that Government lacks, as a standing and coordinated capability, the means to model interacting risks and dependencies; forecast demand, capacity and resource requirements; optimise the allocation of scarce resources; model supply chains,

transport and infrastructure dependencies; evaluate strategies under uncertainty; run scenario and contingency analysis; identify bottlenecks and cascading failures; and translate national risk assessments into quantitatively supported operational decisions. The same risk covers the State's ability to take such capability in when it's offered, whether as methods, models or ready-to-use technology.

The word *integrated* carries weight. The competences this requires sit in separate Departments, as Parts IV to VI show, and the risk lies as much in the absence of a mechanism that assembles them as in the absence of the models themselves. The risk is distinct from each of the individual risks assessed. It concerns the State's capacity to understand, prioritise and respond to them. This is the principal risk in this submission: the management of every other listed risk depends on it, and because, on the record in Part III, offers of that capacity have not been absorbed.

Cross-sector relevance. A major extreme-weather event can affect electricity and communications, transport networks, healthcare capacity, food and fuel distribution, emergency services, housing and displacement, and public finances at once. A pandemic can constrain healthcare capacity, personnel, procurement, transport, inventories, testing, vaccination, communications and public administration simultaneously. These are networks of dependencies, not independent risks. Resilience requires a chain that runs from risk identification through interaction analysis, prioritisation, resource allocation and operational decision to monitoring and adaptation; analytical capability is relevant at every link.

The capability, by name. None of the capabilities listed above is novel. Each is a named class of method in the discipline, used here in its standard textbook sense:

Capability named in the proposed risk	Method class in which it is established	What the method returns
Model interacting risks and dependencies	System dynamics; network analysis; Markov and Monte Carlo methods	A representation of how a disturbance propagates between sectors
Forecast demand, capacity and resource requirements	Time-series and causal forecasting; demand sensing	Expected load with quantified error, and a record of that error
Optimise the allocation of scarce resources	Linear, integer and stochastic programming	An allocation, with the shadow price of each binding constraint
Model supply chains, transport and infrastructure dependencies	Network flows; facility location; multi-echelon design	The routes, stocks and sites on which service depends
Evaluate strategies under uncertainty	Decision analysis; robust and stochastic optimisation; real options; value of information	A ranking of options, and the worth of learning more before choosing
Scenario and contingency analysis	Discrete-event and agent-based simulation; wargaming; digital twins	Responses tested before the physical event
Identify bottlenecks and cascading failures	Queueing theory; theory of constraints; reliability and fault-tree analysis	The point at which the system saturates or fails first
Translate assessments into quantitatively supported decisions	Prescriptive analytics; decision-support systems	What should be done, in what order, with what resources

Analytics is conventionally divided into three tiers: descriptive (what happened), predictive (what will happen) and prescriptive (what should be done). The public descriptions of the exercises in Part VI sit in the first tier and, with horizon scanning, the second. The third is the tier at which Operations Research operates, and it's the one the Assessment's own description does not name.

Review cycle. Inventory theory distinguishes continuous-review systems, which act when a signal crosses a threshold, from periodic-review systems, which observe at fixed intervals and do not adjust between them. The NSRA, repeated at intervals, is a periodic-review system, and a reasonable one for a register. The record in Part III shows what that design implies: signals that arrive between reviews have no route to action before the next. The discipline's remedy is exception-based management, in which anomalies raise alerts and people intervene on exceptions. It is a standing capability, not a document.

Defence, national security and dual-use. Operations Research originated in military operational analysis, and its contemporary applications include force planning, defence logistics and readiness planning, supply-chain optimisation, capability assessment, maintenance scheduling, deployment planning, and contingency and scenario modelling. As Part I records, comparable defence establishments maintain standing operational-analysis functions. On the public record located for this submission, the Irish public system has not institutionalised this analytical layer as a standing function, and in that respect it appears to be an outlier.

The correspondence between the discipline and Defence responsibilities can be stated function by function. The material prepared in 2012 for the Department of Foreign Affairs (the Conspectus 2012) lists eight solver classes with standing Defence functions:

Solver class	Standing Defence function it serves
Assignment	Allocation of personnel, platforms and assets to tasks; rostering
Conformal Mapping	Coordinate and geometric transformation in surveillance, maritime-area and spatial modelling
Data Conditioning	Preparation and validation of operational and logistics data before modelling
Decision Analysis	Capability comparison and choice under uncertainty
Decision Trees	Sequential contingency planning and procurement option analysis
Facility Layout	Design of barracks, depots, workshops and base infrastructure
Financing Structure Analysis	Financing of capital programmes and procurement
Supply Chain Management	Logistics, stockholding and readiness

The mapping is the author's; it represents a taxonomy, not a specification. Each class corresponds to a standing responsibility of the Department of Defence: logistics, procurement, planning, readiness, infrastructure and capability evaluation. Operations Research is not adjacent to that Department's functions; it is embedded in how they are defined.

The same analytical infrastructure serves civilian and Defence purposes, which makes it a useful example of a dual-use strategic capability in the ordinary sense of serving both; no claim is made that any particular method is subject to export control. The export-control regime nonetheless shows the coordination pattern in practice: Regulation (EU) 2021/821 and the Control of Exports Act 2023 place the competent-authority function for dual-use items with the Minister for Enterprise, Tourism and Employment, while the foreign-policy, international-regime and security-related inputs to an assessment sit with other Departments. The function is statutory and located; the inputs are distributed. The strategic question should not be confined within departmental boundaries.

The institutional pattern is now visible in reverse inside the Department of Enterprise, Tourism and Employment itself: the AI Office supplies the coordinating capacity whose absence is recorded for the older Operations Research corpus and for the dual-use assessment process (see Part VII).

Relationship to existing risks. No existing risk is removed. The assessment would recognise analytical and decision-support capacity as a cross-cutting resilience issue:

Existing strategic risk or function	Where integrated analysis is needed
Armed conflict and terrorism	Readiness, logistics, resource allocation and scenario analysis
Defence procurement and readiness	Readiness forecasting, force design and procurement optimisation
Coalition and interoperability planning	Alignment of capability planning with EU and international partnership frameworks
Hybrid threats	Dependency mapping, resilience modelling and scenario analysis
Infrastructure deficits	Investment prioritisation and network analysis
Protectionism and deglobalisation	Supply-chain and dependency modelling
Pandemics, AMR and zoonosis	Capacity, inventory, demand and distribution modelling
Extreme weather and natural disasters	Evacuation, logistics, resource and infrastructure modelling
Food safety, security and supply	Supply-chain, inventory and transportation analysis
AI and other advanced technologies	Governance, validation and decision-support capability
Energy security	Network, capacity and contingency modelling, including fuel logistics
Cybersecurity and data	Dependency, resilience and risk analysis
CBRN threats	Scenario, logistics, resource and consequence modelling

This submission recommends that the NSRA 2026 recognise insufficient integrated analytical and decision-support capability as a cross-cutting strategic resilience risk, and that Ireland's existing analytical capabilities, and the arrangements that connect them, be considered against that requirement.

IX. The institutional and epistemic risk family

The twenty-four listed risks describe external threats and structural pressures. None describes the State's own capacity to recognise, integrate and act on knowledge about them. That capacity is not a single capability. It is a chain, and each link can fail independently of the others. The risks are stated in the Assessment's conditional register. The record in Part III indicates that the condition they describe is already present; what remains prospective is its consequence.

A. The preparedness chain

Link	The question it asks	Proposed risk	Discipline that supplies the link
Recognise	Can the State tell that a signal is significant when it arrives in an unfamiliar form?	2. Epistemic failure	Forecast monitoring; early-warning indicators; sequential analysis
Understand	Can it model how the risk behaves and interacts with others?	1. Analytical capability (Part VIII)	Simulation; system dynamics; network analysis

Link	The question it asks	Proposed risk	Discipline that supplies the link
Coordinate	Can it assemble competences held in separate institutions?	3. Institutional non-integration	Decomposition; hierarchical planning
Determine	Can responsibility converge on a node able to decide, or to route the matter onward?	4. Convergent responsibility	Decision analysis; queueing and admission control
Detect and correct	Can it find out that the earlier links failed, and change course?	5. Loss of self-correction	Statistical process control; earned-value management; after-action review

A failure at any link compromises those downstream. The most serious is a failure of the final link, because it removes the means of discovering the others. Each link has a named discipline, and the five together are the ordinary contents of the field. Nothing in the chain is unusual. The observation is that no one assembles it. Proposed Risks 6, 7 and 8 (Parts IX.F, IX.G and IX.H) are not further links. They are three concrete places where the chain's failure can be observed in the public record: one in a sector, one in a procurement and one across the Anglo-Irish and North-South boundary.

B. Proposed Risk 2: Epistemic failure, or signals that do not reach determination

> ***Epistemic failure.*** *Ireland may face increased strategic vulnerability if its institutions lack sufficient capacity to recognise, preserve, evaluate, integrate and act upon strategically relevant knowledge, particularly where emerging risks cross disciplinary or departmental boundaries. The risk includes the possibility that administrative processes register the receipt, referral or closure of information without producing substantive evaluation or determination.*

The risk. A State can possess relevant information, expertise and evidence and still fail to benefit from them. The failure lies in four distinctions that institutions can collapse: information from knowledge, knowledge from assertion, assessment from acknowledgement, and determination from administrative closure. A document can be received without being evaluated. A proposal can be acknowledged without being tested. A risk can be identified without entering planning. A review can be declared complete while the underlying question stays undetermined. The resulting pathway runs submission, acknowledgement, referral, deferral, closure. The pathway that should have run is recognition, technical evaluation, integration, determination, corrective action or reasoned refusal. The State can appear to have processed a matter without having processed its substance. A discipline with no named entry point in the consultation or impact procedures, such as Operations Research, is exposed to exactly this pathway.

Status. The pathway is described here as a risk because its consequences lie ahead. The pathway itself is already documented: Part III records its operation across six Departments and twenty-seven months.

The corridor. The missing element is a *protected epistemic corridor*: a reliable route by which significant evidence or expertise reaches an authority able to assess it independently, challenge it, integrate it with other knowledge, and either act on it or give reasons for not doing so. A corridor does not oblige the Government to accept every proposition put to it. It obliges the system to be able to say who considered a matter, on what evidence, under what authority, with what result and for what reasons. Recognition of information is not recognition of its significance. Where no corridor exists, uncertainty itself becomes a ground for inaction instead of an object of assessment.

Why it belongs in the NSRA. The Department's consultation document describes the NSRA as horizon scanning designed to avoid group-think. Horizon scanning is a feedback process; this is meaningful only if signals that emerge can enter evaluation and survive institutional processing. The State has itself recorded the

boundary. The Government's release of March 28th, 2024 on the National Risk Assessment for Ireland 2023 notes that some emerging risks had not reached a level of maturity to be analysed by the assessment methodology, and the associated FUTUREPROOF-IE project addresses horizon scanning and emergent-risk forecasting. The open item is the path by which an unfamiliar signal, particularly one originating outside Government, enters that process and reaches a determination. Artificial intelligence, biotechnology, cyber systems, critical infrastructure, supply-chain disruption, climate-related emergencies and hybrid threats all cross departmental lines. A system that identifies familiar categories well and unfamiliar evidence poorly has an epistemic blind spot exactly where emerging risks arise.

Attention saturation. Recognition degrades under concurrent crises. A precursor can be present, visible and documented, and still be missed because analytical attention is committed to a more salient event at the same moment. Risk assessment also concentrates on the most visible threats; contingencies that attract less attention but bear directly on supply routes, infrastructure and operational capacity receive less examination than their consequences warrant. The remedy is structural. It requires a mechanism that reserves analytical capacity for signals that compete poorly for attention, and a practice of testing the prevailing consensus itself, so that group-think is detected by design and not by hindsight.

In the discipline's terms. Signal detection under uncertainty is a developed field. A sequential probability ratio test accumulates evidence observation by observation against two boundaries and stops, by construction, when one is crossed; its defining property is a stopping rule. Forecast monitoring tracks error and flags model degradation. Early-warning indicators and anomaly detection exist to surface what the prevailing model does not expect, and the Delphi method and scenario planning exist for what the data cannot yet show. The pathway in Part III has accumulation without a boundary: thirteen observations, and no point at which accumulation forces a decision. A system with no stopping rule never decides against acting. It keeps observing.

Public-record anchors.

- Department of the Taoiseach, *NSRA 2026 Public Consultation* (purpose: horizon scanning; avoidance of group-think).
- Department of Defence press release, March 28th, 2024 (National Risk Assessment for Ireland 2023; emerging risks outside the methodology; FUTUREPROOF-IE).
- Dáil Éireann, October 8th, 2020, Q7: the National Security Analysis Centre, established in 2019 within the Department of the Taoiseach to coordinate between State bodies and provide strategic analysis of security threats. A recognition and integration function exists for threats; its counterpart for unconventional, cross-disciplinary evidence is the open item.

C. Proposed Risk 3: Institutional non-integration

> ***Institutional non-integration.*** *Ireland may face strategic vulnerability where critical government functions depend upon cooperation between institutions but lack a clearly defined, tested and accountable mechanism for integrating the relevant knowledge, expertise and decision-making authority. Such vulnerabilities may remain hidden because individual institutions can continue to operate correctly within their mandates while the system as a whole fails to produce a substantive assessment, decision or corrective response.*

The institutional double bind. A State can hold competent officials, functioning Departments and clearly defined remits and still be unable to resolve a strategically important question, because the functions do not connect. Each actor can identify a legitimate reason not to act; no actor is positioned to complete the task as a whole. The sequence is familiar: a matter falls between remits, each institution identifies the boundary of its

own responsibility, the matter is referred elsewhere, the receiving institution identifies a different boundary, and the process continues without determination. No single step looks unreasonable. The failure becomes visible only at the level of the function that Government was required to perform.

Inoperable design. Institutional capacity cannot be inferred from the existence of institutions. A system has a capability only if its parts reliably combine to perform the function required of them. In systems engineering the point is routine: a complex system can contain functioning components and fail at integration, because the fault lies in the interface and not in any component. A governance system that performs a function only if recognition, assignment, integration, determination, escalation, feedback and correction all occur has seven places where the interface can fail, and a failure at any one can leave the process running indefinitely while the problem remains unresolved.

A category error in risk governance. The State organises itself by established institutional categories. Emerging risks do not arrive in them. A single strategic problem can involve technology, defence, foreign policy, infrastructure, public health, economic resilience and emergency planning at once. It's too specialised to address without expertise and too interdisciplinary to sit comfortably with one specialist body. Responsibility is therefore distributed while the obligation to reach a coherent national assessment remains. None of this is an argument against specialisation. Distributed responsibility can increase resilience; the risk arises when distribution is not accompanied by integration.

In the discipline's terms. Large problems are routinely solved by decomposition: a master problem holds the constraints that couple the parts, and each subproblem is solved by the party that holds the relevant data. The Dantzig-Wolfe and Benders methods are the standard examples, and hierarchical planning from aggregate to detail is the managerial equivalent. The central lesson is that coupled subproblems cannot be solved independently and summed, because the coupling is the point. Departments are the subproblems, and the statutory record supplies them in good order: thirteen competent authorities, a duty of cooperation, a single point of contact. It does not name the master. A single point of contact relays; a master problem solves. Multi-tier coordination in industry, with its shared planning, information-sharing arrangements and control towers, exists because the absence of a master is a known failure mode.

Public-record anchors.

- Public Service Management Act 1997, s. 4(1)(e) (each Secretary General is to ensure appropriate arrangements for an effective response to matters pertaining to other branches of the public service) and s. 12 (optional joint ministerial orders assigning cross-departmental functions; the Taoiseach told the Dáil on April 27th, 1999 that none had been made; none has been located in the sources consulted).
- S.I. No. 559 of 2024, regs 7 to 11: thirteen competent authorities, a single point of contact in the Minister for Defence, and a requirement for interdependency analysis with no prescribed method or located modelling capacity (Part V).
- Cabinet Handbook (December 2006) ¶3.8: a requirement that Ministers with a functional interest are heard, which is a requirement of consultation, not of integration (Part IV).
- Regulation of Artificial Intelligence Act 2026 and the co-location, within the Department of Enterprise, Tourism and Employment, of the AI Office and the dual-use competent-authority function under Regulation (EU) 2021/821: a concrete demonstration that the State can create a coordinating mechanism when it chooses to do so, and that the same Department now holds both the new coordinating capacity and the older unresolved corpus. This serves as both a counter-example to the non-coordination thesis and a demonstration of the non-recognition with

regards to Operations Research: the State is fully capable of coordination when the recognition is imposed by EU regulation yet remains paralysed with regards to assessing the domestic frontier (Part VII).

- Export control: the competent-authority function for dual-use items sits with one Minister under Regulation (EU) 2021/821 and the Control of Exports Act 2023, while foreign-policy, international-regime and security inputs sit elsewhere (Part VIII).

D. Proposed Risk 4: Convergent responsibility and executive bottlenecks

> ***Convergent responsibility.*** *Ireland may face strategic vulnerability where responsibility for cross-cutting matters converges on a single office or node that lacks effective interfaces for distributing information, obtaining independent challenge, escalating unresolved questions or routing them onward. In that condition formal agency is high and effective agency is low, and the node can become the point at which a significant signal ceases to propagate.*

Antinomic roles. An office-holder can be subject to legitimate obligations that conflict in practice: executive solidarity and independent vigilance, departmental administration and national security, collective responsibility and responsiveness to representations from the public. The obligations are not contradictory in themselves. The risk arises where the institutional design provides no protected channel through which the conflict can be reconciled, so that substantive disagreement is converted into procedural silence. The concern is the design of the role, not the character or intention of any holder of it.

Centralisation is not integration. Concentrating several responsibilities in one political authority does not integrate the underlying functions. Integration requires working interfaces between institutions, information channels, decision authorities and accountability mechanisms. A system can combine high formal authority, wide access to information, multiple departmental responsibilities and substantial resources with low capacity to resolve a cross-cutting problem. The strategic question is therefore not only who is responsible but what happens when responsibility converges on a node that cannot, or does not, resolve the matter, and what prevents that node from becoming the termination point of a significant signal.

In the discipline's terms. A node on which unlike matters converge is a single server fed by heterogeneous arrivals, and queueing theory is specific about what governs its behaviour. As utilisation approaches capacity, waiting time grows without bound. The standard safeguards are a priority discipline, an admission rule that declines work when the queue is too long, and routing to an alternative server. The theory of constraints adds that the throughput of a system is the throughput of its constraint, and that the first task is to find it. Each safeguard is a published, inspectable rule. Where no such rule is published, the wait of a signal at the node is, from outside, unobservable.

Why the test must be designed from within. Collective responsibility (Article 28.4.2°) and Cabinet confidentiality (Article 28.4.3°) are constitutional features. They also mean that the handling of unresolved cross-cutting matters at that level is not externally observable by default. The assessment is positioned to ask the question from inside Government.

Public-record anchors.

- Constitution, Articles 28.4.2°, 28.4.3° and 28.12.
- Public Service Management Act 1997, s. 12(3) (Ministers retain concurrent power to perform functions they assign).

- Cabinet Handbook (December 2006) ¶3.12 and ¶3.16: unreasonable delay and unresolved differences are escalated from Minister to Secretary General to the Taoiseach, which makes the Taoiseach the terminal node by design.
- S.I. No. 559 of 2024 together with the Government Task Force on Emergency Planning: statutory single-point-of-contact, risk-assessment and strategy functions, and the administrative chair of the coordinating forum, converge in the Minister for Defence. The convergence is a design choice and no criticism is implied. It illustrates that convergence is ordinary, which is why the interfaces that let a convergent node propagate and escalate need to be examined.

E. Proposed Risk 5: Loss of institutional self-correction

> ***Loss of institutional self-correction.*** *Ireland may face strategic vulnerability where the mechanisms intended to detect institutional failure are subject to the same structural limitations as the institutions they oversee. Each institution can then demonstrate that it followed its own procedures while the State, taken as a system, cannot demonstrate that the underlying function was performed. In high-consequence environments this produces a dangerous divergence between the appearance of preparedness and actual preparedness.*

The nature of the risk. The risk is to the State's capacity to correct itself. A constitutional democracy depends on institutions that can receive information, recognise evidence, identify failure, assign responsibility, correct defective decisions and learn from the result. If those functions become mutually avoidant, the system loses the property that matters most in a crisis: the ability to discover that it has made a bad decision, understand why, and change course. No claim is made that Ireland's constitutional order has ceased to function. The proposition concerns risk.

Institutional self-confirmation. A system that measures the completion of its processes, and not the completion of the function those processes serve, can report success while failing. Correspondence is received, a reference number is issued, a Department replies, the matter is referred, a committee considers its jurisdiction and a file is closed. Each event is procedurally valid. None establishes that the underlying question was answered. The result is procedural resilience without functional resilience: every node reports that it has performed its local function while the collective obligation remains unmet. The condition requires neither bad faith nor incompetence. It arises from the interaction of ordinary procedures.

In the discipline's terms. Statistical process control exists to close exactly this loop: monitor, detect, diagnose, correct. Its first discipline is to plot the quantity that matters. A control chart that records how many inspections were performed, and not what they found, is a control chart in form only. Earned-value management makes the same point for projects: progress is measured by work accomplished against plan, not by money spent or effort expended. On that measure, and on the record located for this submission, the thirteen acknowledgements in Part III are cost and the determinations are earned value; the second series is empty. Risk management is, in the discipline's own classification, a cybernetic activity: indicators are monitored, mitigation is triggered at thresholds, and effectiveness is observed and fed back. The Assessment is risk management. Whether the loop is closed at the level of the State's own handling of risk is the question the fifth risk asks.

Why ordinary accountability can miss it. Existing accountability mechanisms characteristically ask whether a particular actor complied with a particular procedure. They are less well equipped to detect a failure at the interface between legitimate actions by different actors. The system requires a question at the level of function: not *did each institution perform its assigned procedure*, but *did the State, taken as a system, accomplish the function required.*

The constitutional dimension. A representative system depends on a chain that runs from public concern or evidence, through representation, recognition, responsible consideration and determination, to action or reasoned refusal, accountability and correction. The administrative appearance of each stage can be completed without its substance. The relevant capability is not universal responsiveness; Government need not accept every warning or proposal. It is *traceable determination*: for any significant matter repeatedly represented to Government, an identifiable point at which the system can say who considered it, what evidence was considered, which authority was responsible, what was determined, on what reasons, and what mechanism exists for review or correction. A gap between having responsibility and being able to exercise it, if persistent, weakens the practical standing of representative institutions even where no law is breached.

Institutional memory. Self-correction requires memory. Where responsibility moves between Departments, Ministers and Governments, an unresolved matter can be treated as a first encounter and the cycle of superficial closure restarts. The Cabinet Handbook provides for one standing device for cross-cutting matters, the Cabinet Committee, established by Government decision where a remit has a cross-cutting dimension (¶2.24 and Appendix VIII). Committees stand dissolved at the end of every Government's term, and the Handbook restricts disclosure of committee information beyond existence, purpose and membership. A cross-departmental analytical team can therefore be re-established with each Government. If one exists, it's neither standing across Governments nor open to inspection.

Public-record anchors.

- Cabinet Handbook (December 2006), ¶2.24 and Appendix VIII (dissolution at the end of each Government's term; limits on disclosure).
- Public Service Management Act 1997, s. 4(1)(b): strategy statements every three years, a periodic review mechanism that does not trace a signal to determination.
- Part III of this submission: thirteen acknowledgements from six Departments since June 2024, none addressing the cross-departmental question.

F. Proposed Risk 6: Geoeconomic exposure of digital infrastructure

> ***Geoeconomic exposure of digital infrastructure.*** *Ireland may face strategic vulnerability where a concentrated base of digital infrastructure (data centres, cloud services, the electricity supply that sustains them and the technology flows that pass through them) becomes an object of geoeconomic competition, while the instruments through which the State manages that competition are administered separately and are not assessed together.*

Concentration. The Central Statistics Office reports that data centres consumed 22% of metered electricity in Ireland in 2024, up from 5% in 2015, and its 2025 release reports 23%. Large energy users, a category that includes the larger data centres, accounted for a third of metered consumption in 2025. Energy security (Risk 21), cybersecurity and data (Risk 23) and artificial intelligence (Risk 20) meet in this one asset class. No listed risk owns the intersection.

Instruments. The State holds several instruments for managing geoeconomic competition in this sector, each with its own administrator and its own test:

- *Export controls.* Regulation (EU) 2021/821 and the Control of Exports Act 2023 place the competent-authority function for dual-use items with the Minister for Enterprise, Tourism and Employment.
- *Investment screening.* The Screening of Third Country Transactions Act 2023, in operation since January 6th, 2025, requires notification of covered acquisitions by non-EU, non-EEA and non-Swiss investors

in critical infrastructure and critical technologies, which the Department's guidance defines by reference to the dual-use items in Annex I of the same Regulation. The Department's first annual report records 26 transactions screened in 2025, none prohibited, of which 18 concerned critical infrastructure and four concerned critical technologies and dual-use items.

- *Critical-entities resilience.* S.I. No. 559 of 2024 provides for sectoral competent authorities and the Minister for Defence as a single point of contact; digital infrastructure is a sector of the underlying Directive.
- *Energy connection and planning policy* for large users, administered by the energy regulator and the planning system.

The integration gap. The first two instruments sit with one Minister and the third with another. Their combined effect on a single sector is the question that matters to resilience. Public descriptions of the three exercises named in Part VI do not describe an assessment of that cumulative exposure. The dual-use definition illustrates the interdependence: the same Annex I list is the spine of export control and of investment screening, so a classification made under one regime has consequences under the other. This is Proposed Risk 3 (institutional non-integration) in a concrete, verifiable form.

In the discipline's terms. Interdependent infrastructure is a network, and the questions that matter about a network have standard answers: which single node or link, if lost or restricted, disconnects the most (the minimum-cut problem); how a disturbance propagates between the layers of a coupled system (supply-chain and power-network disruption models); how a grid behaves under contingency (optimal power flow, load forecasting, digital twins). Competition over a shared asset adds game theory and mechanism design. All of these require one graph. The instruments listed above each hold a fragment of it: licences, notified transactions, designated entities, connection agreements. None of them assembles the graph, and the public record located for this submission does not show that it is assembled elsewhere.

Why it belongs in the NSRA. Geoeconomic confrontation is the leading global risk in the World Economic Forum's *Global Risks Report 2026*, and Ireland hosts a concentration of the infrastructure that such competition targets. The risk is distinct from Risk 21 because it concerns the political and regulatory exposure of the asset, not the adequacy of supply.

Public-record anchors.

- Central Statistics Office, *Data Centres Metered Electricity Consumption* (2024 and 2025 releases).
- Screening of Third Country Transactions Act 2023; Department of Enterprise, Tourism and Employment, first annual report under the Act (January 6th to December 31st, 2025).
- Control of Exports Act 2023; Regulation (EU) 2021/821, Annex I.
- S.I. No. 559 of 2024.
- World Economic Forum, *Global Risks Report 2026* (January 14th, 2026).

G. Proposed Risk 7: Delivery of critical national systems

> ***Delivery of critical national systems.*** *Ireland may face strategic vulnerability where critical operational systems are procured and funded in dependent stages without a standing practice of demonstrating the foundational stage at operational scale before later-stage expenditure is committed, and where the contracting body, the approving authority*

> *and the sponsoring Department each hold part of the decision. Failure may then surface late, as write-down, delay and continued reliance on legacy systems, and not early as a managed stop.*

The case. In May 2026 Iarnród Éireann wrote down approximately €50 million on its Traffic Management System, a component of the National Train Control Centre project, under a contract dating from 2020. The Oireachtas heard that testing in mid-2025 identified systemic issues, that the supplier's own safety assessor found the software not ready for service, and that completion is not expected before 2030. The contract has since been terminated. The supplier disputes responsibility and the National Transport Authority has commissioned an independent review. Iarnród Éireann is the contracting body, the National Transport Authority is the approving authority for the capital expenditure, and the Department of Transport is the sponsoring Department. No finding is made here, and no breach of any procurement rule is alleged.

Why it belongs in the NSRA. In a sequentially dependent system, later stages have value only if earlier ones work. That property can be analysed before money is committed, through stage dependency, cost accumulation and risk allocation. These are routine methods of Operations Research and systems engineering. Existing approval frameworks apply to such projects. Whether they test demonstrated function at operational scale before later-stage release is the question for the assessment. The risk touches fiscal sustainability (Risk 6), infrastructure deficits (Risk 7) and rail expansion (Risk 16), and it is Proposed Risk 3 in a concrete form: three bodies each hold part of the decision, and none holds the whole.

In the discipline's terms. A programme whose later stages depend on earlier ones is a serial system, and its prospect of delivery is bounded by the weakest stage. Stage-gate review with an explicit option to abandon is the standard way of managing that structure: real-options methods value the right to stop, and the value of that right is realised only if the gate tests the quantity that matters, demonstrated function at operational scale, and not the completion of the activity. Earned-value management makes the same distinction for cost: spend recorded against a stage is not progress unless the stage's output works. Critical-path analysis identifies the stages on which the whole depends, and fallback planning identifies, before termination is necessary, the legacy capability on which the service continues to rely.

Public-record anchors.

- Dáil Public Accounts Committee hearing reported by the *Irish Times* and RTÉ, May 21st, 2026 (the write-down); RTÉ, June 10th, 2026 and July 18th, 2026 (testing and completion timeline).
- *Irish Times*, September 8th, 2026 (termination of the contract; the supplier's account; prospect of litigation).
- Public Accounts Committee, July 16th, 2026 (National Transport Authority accepts Iarnród Éireann's recommendation to end the supplier's role).

H. Proposed Risk 8: Fragmented Anglo-Irish and cross-border coordination capacity

> ***Fragmented Anglo-Irish and cross-border coordination capacity.*** *Ireland may face strategic vulnerability where risks that cross the land boundary with Northern Ireland, or that depend on coordination with the United Kingdom more generally, are managed through institutions each competent for part of the relationship, without a standing capacity to model how a disturbance in one domain propagates through the others. The vulnerability is the external counterpart of Proposed Risk 3: the same absence of an assembling mechanism that affects coordination between Irish Departments also governs coordination across the boundary.*

The architecture. Strand Two of the Belfast/Good Friday Agreement (1998) established the North/South Ministerial Council, bringing the Government and the Northern Ireland Executive together to oversee twelve

agreed areas of cooperation: six implemented by common policy, agreed in the Council and applied separately in each jurisdiction (agriculture, education, environment, health, tourism and transport), and six implemented directly through North/South Implementation Bodies. Strand Three established the British-Irish Council and the British-Irish Intergovernmental Conference, bringing in the devolved administrations of Scotland and Wales and the Crown Dependencies. Since 2007 the all-island Single Electricity Market, now the Integrated Single Electricity Market, has coupled electricity supply and pricing across the boundary. Since February 2023 the Windsor Framework has governed the movement of goods between Great Britain and Northern Ireland, with consequences for the onward movement of goods, including dual-use and controlled items, into the State. The common travel area predates the State itself. Each institution is competent within its own domain. The public record located for this submission does not show a standing capacity that models how a disturbance in one, an electricity-market contingency, a goods-movement restriction, a health-system surge affecting cross-border patient flows, propagates into the others, or into the risks already listed in the Assessment.

Why it belongs in the NSRA. Regulation 11(3) of S.I. No. 559 of 2024 already requires the national risk assessment to account for risks of a cross-sectoral *or cross-border* nature, so the instrument anticipates exactly this category without prescribing how the analysis is to be performed (Part V.D). The export-control thread running through this submission (Part VIII) supplies a concrete instance: a dual-use or controlled classification made by the competent authority in the State has a distinct, and not necessarily aligned, counterpart on the other side of the land boundary under the Windsor Framework's separate arrangements, and no public mechanism identified for this submission models the combined effect. The same is true of electricity: a contingency in the all-island market engages the regulators, grid operators and Departments of two jurisdictions at once, and Risk 21 (Part VI, Q1) already identifies domestic fuel-logistics exposure without extending the analysis across the boundary. Where a risk is transboundary by construction, an assembling mechanism confined within the State cannot, on its own terms, complete the analysis the Regulation requires.

In the discipline's terms. A boundary that two administrations manage jointly is a classic instance of a decentralised, multi-agent system: each side optimises locally, under its own objectives and constraints, and the combined outcome can diverge from what either side would choose were the coupling modelled explicitly. Coordination games and mechanism design characterise when decentralised optimisation converges on a jointly efficient outcome and when it does not; the network and dependency analysis that Risk 6 (Part IX.F) applies within the State extends without modification across a jurisdictional boundary once the relevant data are shared. None of this implies that the North/South or British-Irish institutions are defective in their own terms. It implies that their outputs are inputs to a wider analysis that, on the record located for this submission, no body is shown to perform.

Limits. No claim is made about the merits, legal status or political sensitivity of any Anglo-Irish or North-South arrangement, and nothing in this risk touches the constitutional status of Northern Ireland. The risk is confined to the analytical and coordination capacity question that is this submission's subject throughout, applied to the one boundary across which Ireland's risk environment is, by the State's own statutory instrument, already required to be assessed.

Public-record anchors.

- Belfast/Good Friday Agreement (1998), Strand Two (North/South Ministerial Council, twelve areas of cooperation, six North/South Implementation Bodies) and Strand Three (British-Irish Council; British-Irish Intergovernmental Conference).
- S.I. No. 559 of 2024, reg. 11(3): the national risk assessment must account for risks of a cross-sectoral or cross-border nature (Part V.D).

- Windsor Framework, in operation from February 2023, governing the movement of goods between Great Britain and Northern Ireland.
- Single Electricity Market / Integrated Single Electricity Market, operating on an all-island basis since 2007.
- Regulation (EU) 2021/821 and the Control of Exports Act 2023 (Part VIII), for the export-control instance of the cross-border gap.

I. Resilience tests for the assessment

The assessment could establish the following. They are tests, not allegations.

1. How Government identifies strategically significant knowledge that falls between established departmental categories.
2. What mechanism evaluates evidence or expertise that is unfamiliar, interdisciplinary or cross-cutting, without requiring it first to conform to an existing departmental category.
3. How Government records distinguish administrative receipt of information from substantive evaluation of it, and whether any measure compares determinations with acknowledgements.
4. What identifies a matter that has passed through several institutions without receiving a substantive determination.
5. How institutional memory is preserved when responsibility moves between Departments, Ministers and Governments.
6. Who is responsible for identifying systemic failure when every participating institution can show that it performed its individual procedural function.
7. What feedback path allows evidence of an unresolved or mishandled strategic risk to travel upward until corrective action occurs.
8. Whether the State periodically tests its own capacity to detect institutional blind spots. A practical instrument is a *signal-traversal test*: a documented, anonymised cross-boundary signal is traced from entry to determination, and the result is reported to the Government Task Force on Emergency Planning and into the NSRA process.
9. **Horizon allocation.** What share of Government's strategic planning effort addresses structural change over a decade-long horizon, as against short-run volatility, and whether the State holds a standing, shared view of how the international order on which it depends may change.
10. **Consensus testing.** Whether the assessment tests its own prevailing consensus through red-team review or competing hypotheses.
11. **Scenario exercises.** Whether Departments run cross-departmental scenario exercises supported by quantitative models of resource, transport and supply dependencies, and whether the findings return to the assessment. The fuel disruption that began on April 7th, 2026 (Q1 above) is a public-record case against which such an exercise could be run.
12. **Concentration points.** Whether irreplaceable national dependencies, such as single suppliers, corridors, interconnections and facilities, are identified through cross-departmental assembly and not left to sector-by-sector identification. S.I. No. 559 of 2024 assigns identification of critical entities to sectoral competent authorities; dependencies that cross sectors are visible only in aggregate. The domestic refinery and the two depots through which, on the industry body's account, roughly half of national fuel supply passes are an instance.

13. **Stopping rule.** Whether any matter represented to Government repeatedly has a defined point at which accumulated representations require a determination, a reasoned refusal or a referral to a named owner.
14. **Staged release.** Whether release of staged expenditure on dependent systems is conditional on demonstrated performance of the foundational stage at operational scale, and whether the condition is recorded.
15. **Fallback path.** Whether major systems carry a documented fallback path developed before termination becomes necessary.
16. **Defence procurement.** Whether Defence procurement frameworks include a formal analytical stage, covering readiness forecasting, capability comparison and supply-chain optimisation, as a condition of major expenditure decisions, and whether that stage is independently validated.

A mature risk-assessment system should be able to meet each of these.

J. An illustration, and its limit

A documented Irish record illustrates the failure mode. Operations Research material presented to State bodies since 2006 has generated a series of acknowledgements across successive Governments, Ministers and Departments (Part III), with the same output each time: acknowledgement without a determination on the national-capability question. No consolidated, documented technical assessment has been identified on the departmental record, and the discipline has no procedural entry point through which one would be required. The record is offered as evidence that the pattern can occur and remain difficult to detect, not as an argument about the merits of any individual matter. Its significance for the NSRA is the possibility that the same failure mode operates elsewhere, in areas where no one is placed to notice. The present submission is a further instance and a further test. It enters the one channel the Department has opened for unfamiliar evidence, on the Department's invitation. Its passage is observable, and the Assessment is placed to observe it.

X. What the assessment should examine

The assessment could examine nine matters:

1. where analytical and decision-support capabilities currently reside across Government, Department by Department;
2. whether a standing mechanism assembles those competences for risk-interaction analysis, including arrangements made under section 4(1)(e) of the Public Service Management Act 1997 and any order made under section 12;
3. whether methods and data can be integrated across departmental boundaries;
4. whether critical analytical capability is available during major emergencies, including through the Government Task Force on Emergency Planning and the National Emergency Coordination Group;
5. whether sovereign or appropriately controlled analytical capability exists for high-consequence functions;
6. whether analytical outputs are incorporated into contingency planning and Government decisions, including whether memoranda for Government should address cross-sector interdependence where a proposal bears on a listed risk, which neither the impact headings of Cabinet Handbook ¶3.4 nor the Regulatory Impact Analysis model in Appendix III currently asks for, and whether any Cabinet

Committee or cross-departmental team holds a standing remit for risk interaction that survives a change of Government, to the extent that can be disclosed;

7. the relationship between the NSRA, the National Risk Assessment for Ireland and the Defence-led assessment of Departmental capabilities, and whether the three share an analytical base;
8. whether adequate validation and human oversight exist for high-consequence analytical and AI-assisted decisions; and
9. whether the cross-border risk analysis that reg. 11(3) of S.I. No. 559 of 2024 requires is in fact performed, and if so, by what body, drawing on which of the North/South Ministerial Council, the British-Irish Council, the Single Electricity Market and the Windsor Framework's goods-movement arrangements, or whether each is left to assess its own domain without a shared model of how a disturbance in one propagates into the others.

Two proportionate forms of enquiry follow, either of which could be undertaken without a policy change. The first is a defined, time-bound assessment of existing State analytical capability relevant to Defence and emergency planning. The second is a review by an appropriate committee or interdepartmental group to clarify where responsibility for such assessment lies, to examine the processes applied to date, and to report. Each concerns procedural completeness and administrative assurance. Each would, it is submitted, be limited in scope and capable of being met from existing resources.

None of this requires a new Department, agency or technology programme. The first requirement is that the capability question be recognised and assessed.

XI. What this submission does not prescribe

This submission does not propose that the NSRA adopt a particular commercial technology, select a supplier, or prescribe an Operations Research methodology. The method classes named in Parts VIII and IX locate the capability by naming what the discipline contains. They are a taxonomy and not a specification. It does not propose that departmental risk management be replaced, that AI be treated as preferable to existing analytical methods, or that any conclusion be reached in advance about an individual technology or proposal. It does not prescribe an institutional form. Statute, procedure and precedent already offer several: an order under section 12 of the 1997 Act, a standing function within the Government Task Force on Emergency Planning, a Cabinet Committee remit, or a coordinating office on the model of the AI Office of Ireland. The choice among them can be made on the evidence the assessment gathers. The risks in Part IX are structural. They do not attribute bad faith or incompetence to any office-holder, and they do not ask the NSRA to reach a conclusion on any individual matter, including the Traffic Management System and the fuel disruption, which are cited as public-record cases and not as findings. The exclusion described is procedural and is not a finding on merit. The assessment can adopt the proposed risks, in whole or in part, without a finding about any institution's conduct.

XII. The open question

The documentary record shows a proposition presented since 2006 and acknowledged across six Departments, without being assessed as a question of national capability. The procedural record shows an executive that requires consultation, an architecture that does not require integration and, on a Senator's account, committees that do not replicate even the consultation. The statutory record shows a risk-assessment regime that requires interdependency analysis and presupposes the capacity to perform it. The domestic precedent shows a State able to build coordination when it decides to. The delivery record shows a critical operational system advanced through dependent stages to a write-down of approximately €50 million, with the decision divided among a contracting body, an approving authority and a sponsoring Department. One question follows from the five, and it is the question this submission asks the NSRA to carry:

> ***Does Ireland possess a standing, integrated capability, both analytical and institutional, to recognise, understand and act on complex, interdependent national risks, together with a mechanism that detects and corrects failures of that capability?***

If the answer is yes, the assessment should be able to identify where the capability sits, how its competences are assembled, and what tests it. If the answer is no, the gap is itself a strategic resilience risk, and the NSRA is the instrument through which it is recorded.

Sources

Constitutional, statutory and EU instruments

- Bunreacht na hÉireann, Articles 28.4.1° to 28.4.3° and 28.12.
- Public Service Management Act 1997 (No. 27 of 1997), ss. 4(1)(b), 4(1)(e) and 12: irishstatutebook.ie/eli/1997/act/27
- European Union (Resilience of Critical Entities) Regulations 2024 (S.I. No. 559 of 2024), regs 2, 7, 8, 9, 10 and 11: irishstatutebook.ie/eli/2024/si/559
- Regulation of Artificial Intelligence Act 2026 (signed July 21st, 2026).
- Directive (EU) 2022/2557 (resilience of critical entities); Decision No 1313/2013/EU on a Union Civil Protection Mechanism, Articles 4 and 6; Regulation (EU) 2021/821; Control of Exports Act 2023; Screening of Third Country Transactions Act 2023.

Procedure and administration

- Department of the Taoiseach, *Cabinet Handbook* (December 2006), ¶¶ 1.1 to 1.2, 2.24, 3.3 to 3.4, 3.8 to 3.12, 3.15 to 3.16, Appendix III and Appendix VIII: gov.ie/en/publication/05c2e2-cabinet-handbook
- Dáil Éireann, April 27th, 1999 (orders under section 12 of the 1997 Act): oireachtas.ie/en/debates/debate/dail/1999-04-27/4
- Dáil Éireann, March 25th, 2004 (emergency planning): oireachtas.ie/ga/debates/question/2004-03-25/2/
- Dáil Éireann, February 7th, 2013, Q29 (lead-Department principle): oireachtas.ie/en/debates/question/2013-02-07/29/

- Dáil Éireann, November 15th, 2022, Q313 (National Emergency Coordination Group): oireachtas.ie/ga/debates/question/2022-11-15/313
- Dáil Éireann, October 8th, 2020, Q7 (National Security Analysis Centre): oireachtas.ie/ga/debates/question/2020-10-08/7/
- Dáil Éireann, May 16th, 2024, Q44 (National Risk Assessment for Ireland 2023): beta.oireachtas.ie/en/debates/question/2024-05-16/44/
- gov.ie, *An Taoiseach* (role description); Office of Emergency Planning, *Critical Entities Resilience (CER) Regulations*; Department of Defence press release, March 28th, 2024 (National Risk Assessment for Ireland 2023 and FUTUREPROOF-IE).
- Department of the Taoiseach, *National Strategic Risk Assessment 2026: Public Consultation* (published September 11th, 2026): gov.ie/en/department-of-the-taoiseach/consultations/national-strategic-risk-assessment-public-consultation-2026/
- Department of the Taoiseach, *National Risk Assessment 2024: Overview of Strategic Risks* (September 2024).

Economic and statistical sources

- World Economic Forum, *Global Risks Report 2026* (January 14th, 2026).
- Central Statistics Office, *Data Centres Metered Electricity Consumption 2024* (June 10th, 2025) and the 2025 release.
- Department of Enterprise, Tourism and Employment, first annual report on the screening of third-country transactions (covering 2025).

Parliamentary record

- Seanad Select Committee on EU Scrutiny and Transparency, Tuesday, March 3rd, 2026, Any Other Business (Senator Michael McDowell, in discussion with Senator Alice-Mary Higgins).

Artificial Intelligence comparator

- Department of Enterprise, Tourism and Employment, *General Scheme of the Regulation of Artificial Intelligence Bill 2026* (February 2026), Heads 9, 15 and 34; departmental news release, August 5th, 2026; Matheson, *Irish Artificial Intelligence Act 2026 signed into law*; William Fry, *Irish AI Office Loses Market Surveillance Role* (June 30th, 2026).
- Herbert A. Simon, "Two Heads Are Better than One: The Collaboration between AI and OR", *Interfaces 17(4)*, 1987

Operations Research: technical demands

- Educational Testing Service, *GRE Guide to the Use of Scores*, Table 3B: GRE General Test Interpretive Data by Graduate Major Field (seniors and non-enrolled college graduates tested July 1st, 2022 to June 30th, 2025). Operations Research: 436 test takers, Quantitative Reasoning mean 163, standard deviation 6 — ets.org/pdfs/gre/gre-guide-table-3b.pdf
- Educational Testing Service, *GRE Guide to the Use of Scores*, Table 3A: Percentage Distribution of Scores Within Intended Broad Graduate Major Field (same population and period) — ets.org/pdfs/gre/gre-guide-table-3a.pdf

- Analysis Function, UK Civil Service, *Government Operational Research Service* — analysisfunction.civilservice.gov.uk/government-operational-research-service
- U.S. Army, *An introduction to uniformed operations research* (Functional Area 49, Operations Research/Systems Analysis) — army.mil/article/140106

Fuel logistics (April 2026)

- RTÉ News, April 10th, 2026 and April 13th, 2026 (blockades of the Whitegate refinery and the Foynes and Galway depots; National Emergency Coordination Group; Fuels for Ireland statements on forecourt outages).
- *Irish Times*, CEPA and Al Jazeera coverage of the blockades, April 7th to April 14th, 2026.

Delivery of critical national systems

- *Irish Times*, May 21st, 2026 (Public Accounts Committee; write-down of €50 million) and September 8th, 2026 (termination; dispute with the supplier).
- RTÉ News, May 21st, 2026, June 10th, 2026 and July 18th, 2026.
- Public Accounts Committee, July 16th, 2026 (National Transport Authority statement on the supplier's role).

Operations Research in Defence

- *Conspectus 2012* (material prepared for the Department of Foreign Affairs); Annex, *Strategic Relevance of Operations Research to Defence and National Security*, prepared for the Clerk of the Committee on Defence and National Security.

Anglo-Irish and cross-border architecture

- Agreement Reached in the Multi-Party Negotiations (the Belfast/Good Friday Agreement), April 10th, 1998, Strand Two and Strand Three.
- North/South Ministerial Council, *Areas of Co-Operation* (12 agreed areas; 6 North/South Implementation Bodies): northsouthministerialcouncil.org
- Windsor Framework (agreed February 27th, 2023), governing the movement of goods between Great Britain and Northern Ireland.
- Single Electricity Market Committee / Integrated Single Electricity Market, operating on an all-island basis since 2007.
- European Union (Resilience of Critical Entities) Regulations 2024 (S.I. No. 559 of 2024), reg. 11(3) (Part V.D).